



Research Question

LLM training data is overwhelmingly English: ~92.7% of GPT-3’s tokens, ~89.7% of LLaMA2’s, with newer models like OLMo 3 filtering for English only. We ask:

Does the language of mathematical problem presentation affect LLM performance?

Mathematical reasoning is an ideal probe of cross-lingual ability: the concepts are language-independent, only the framing changes. **Icelandic** (~370K speakers, four cases, three genders, rich morphology) is a low-resource testbed that shares Germanic roots with English.

Parallel Sample

Algebra (Level 3) Answer: B

ICELANDIC

Rögvaldur og vinir hans fjórir eiga að meðaltali 220 kr., en Rögvaldur sjálfur á 380 kr. Hve mikið eiga vinirnir fjórir að meðaltali?

ENGLISH

Rögvaldur and his four friends own on average 220 kr., but Rögvaldur himself owns 380 kr. How much do his four friends own on average?

(A) 160 kr. (B) 180 kr. (C) 220 kr. (D) 300 kr.

Number Theory (Level 4) Answer: B

ICELANDIC

Hver er afgangurinn þegar 11 er deilt upp í $10^{10} + 1$?

ENGLISH

What is the remainder when 11 divides $10^{10} + 1$?

(A) 0 (B) 2 (C) 5 (D) 7

Resource: STAK

847
problems

41
years

4
domains

10
difficulty levels

Each problem is paired with a Gemini-3-Flash machine translation into English. **100 random translations** were manually verified by a native Icelandic speaker with professional English proficiency. Semantic faithfulness confirmed.

Methodology

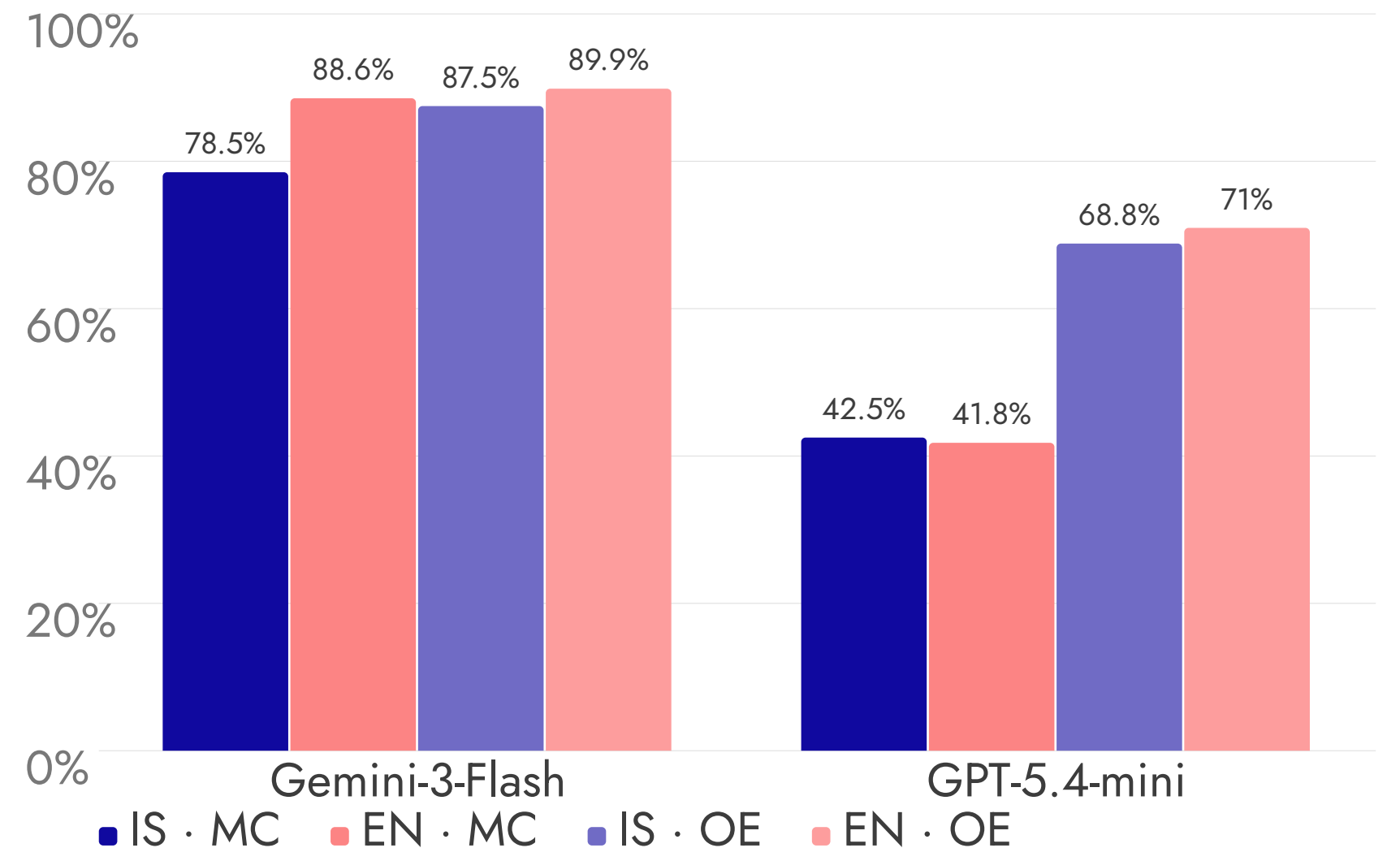
Models. Gemini-3-Flash-Preview (Google) and GPT-5.4-mini (OpenAI), evaluated via OpenRouter with identical prompts.

Dual modes. Each problem is run with multiple-choice options (MC) and as open-ended (OE), to test whether choice scaffolding mitigates or exacerbates language effects.

Three-judge quorum. Correctness determined by majority vote of independent judges from three providers (Gemini, GPT, Claude). 97.6% unanimous agreement across 6,776 evaluations.

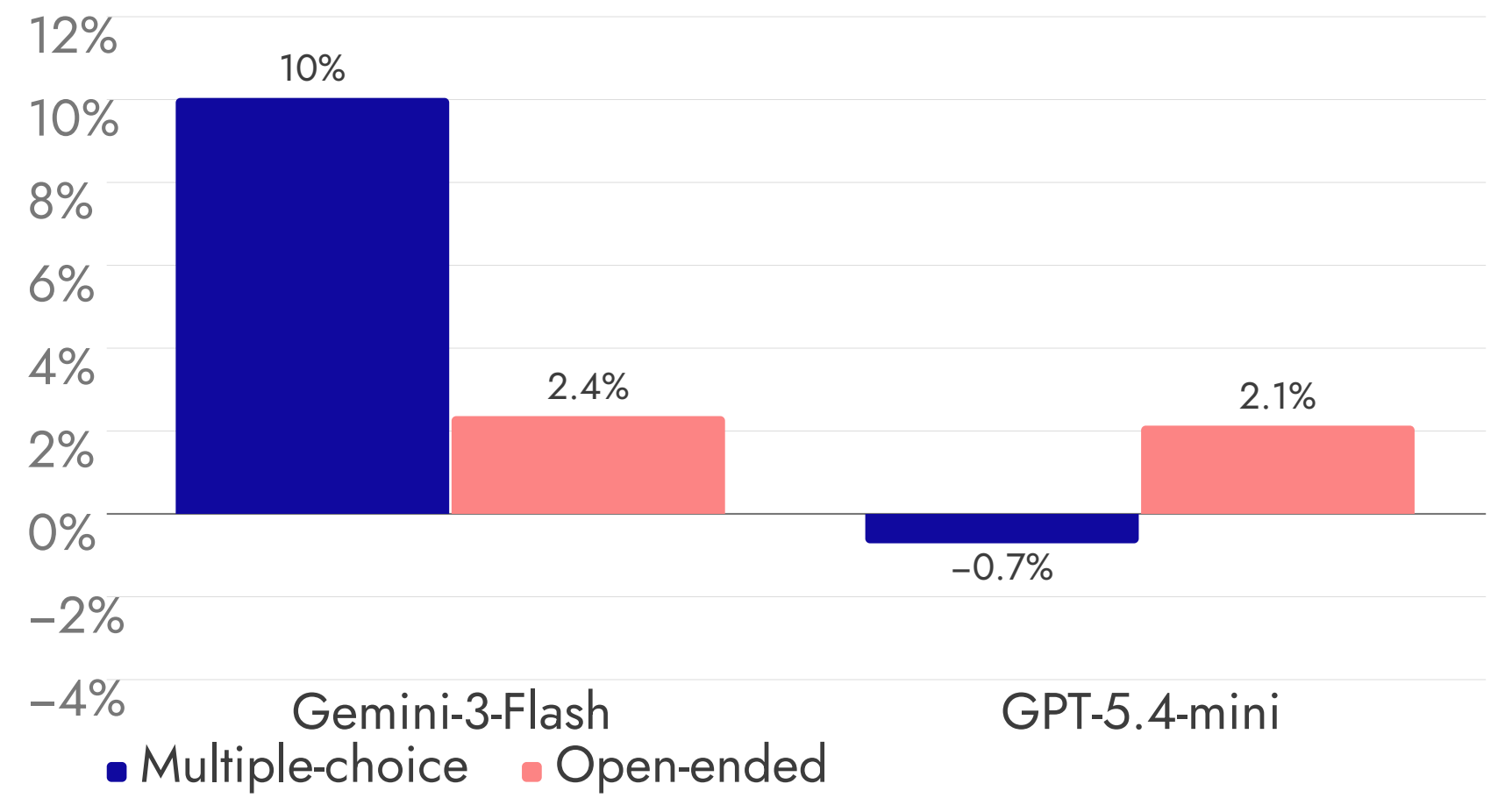
Statistical tests. McNemar’s test for paired binary outcomes; bootstrap CIs (10,000 resamples).

Overall Accuracy



Gemini-3-Flash dominates GPT-5.4-mini in every condition. Both models do better in English. The OE/MC gap is the surprise: dramatic for GPT-5.4-mini, modest for Gemini.

Language Effect (EN – IS)



Gemini-3-Flash: significant English advantage in both modes (MC $p < 0.001$, OE $p = 0.002$).
GPT-5.4-mini: no significant language effect (MC $p = 0.703$, OE $p = 0.098$).

Language effects are model-dependent. One model is sensitive to the presentation language, the other is not.

Discordant Pairs (McNemar)

Per-problem agreement between English and Icelandic versions reveals where each model loses or gains:

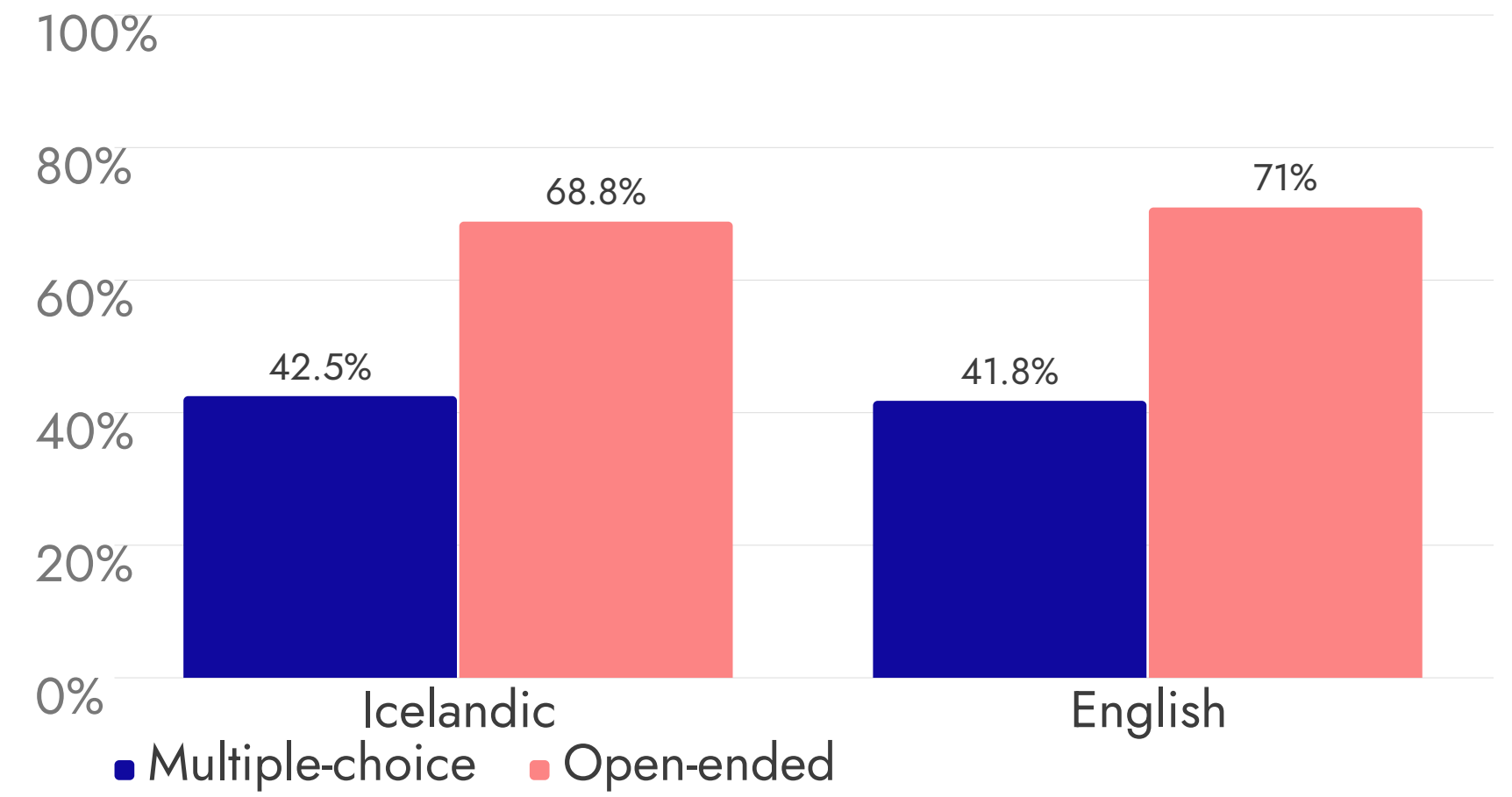
Model / Mode	Both ✓	EN only	IS only	Both ✕	<i>p</i>
Gemini · MC	639	111	26	71	< 0.001
Gemini · OE	730	31	11	75	0.002
GPT · MC	233	121	127	366	0.703
GPT · OE	533	68	50	196	0.098

For Gemini, items right in English but wrong in Icelandic outnumber the reverse 4x. For GPT-5.4-mini the two columns are roughly balanced.

Cross-Generational

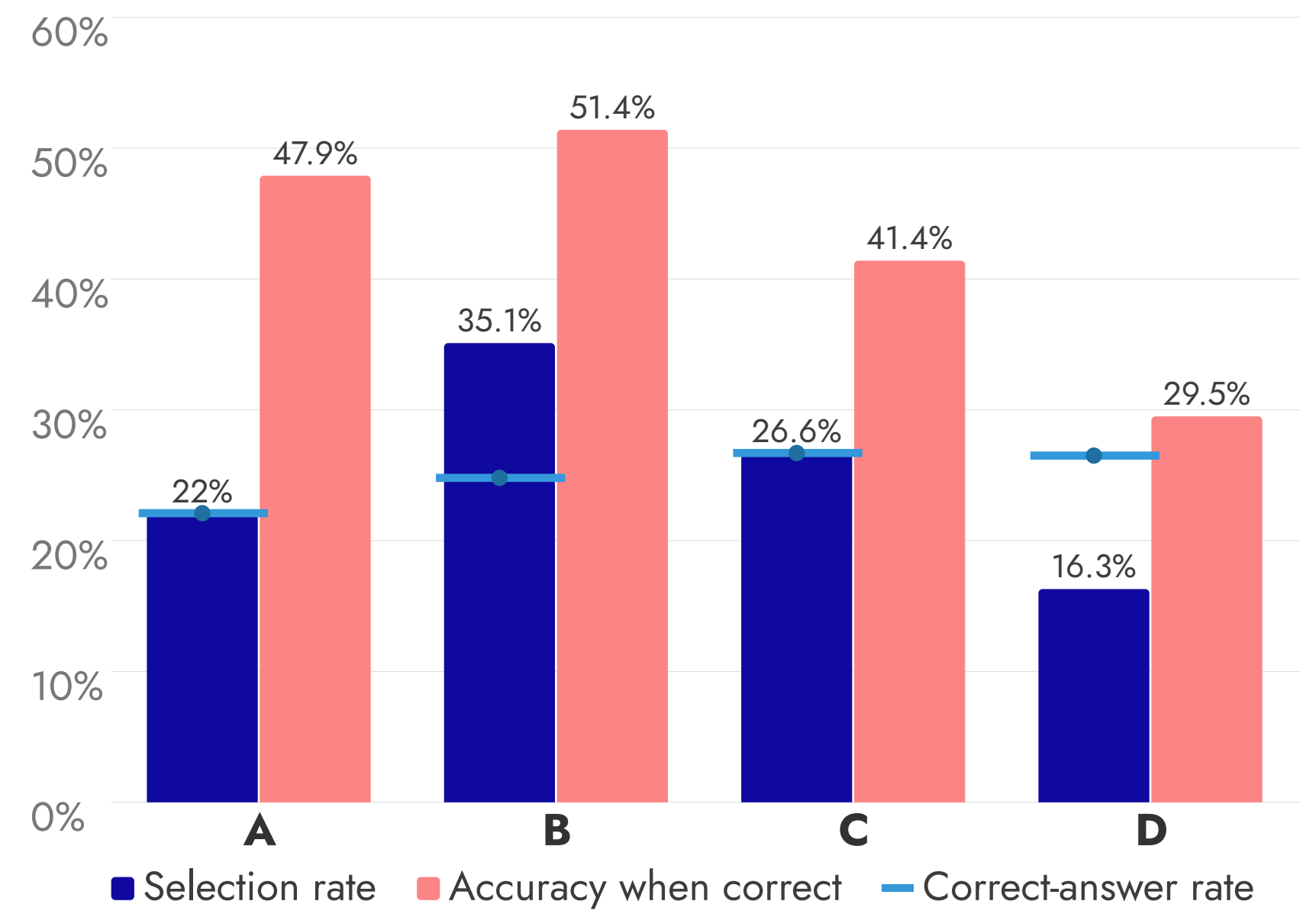
GPT-5.2 shows the same pattern: 49% MC vs. 80% OE (≈ 31 pp deficit) with similar position bias (over-selects B and C, under-selects A). The deficit persists across two OpenAI generations, which points to an architectural or training cause rather than a single-release issue.

Surprise: GPT’s MC Deficit



GPT-5.4-mini drops ≈ 27 pp when answer choices are present. That reverses the typical 25 pp drop seen when MC scaffolding is removed (Myrzakhan et al. 2024). MC format systematically **disadvantages** this model.

Why? Option Position Bias



GPT-5.4-mini **over-selects B** (35.1% vs 24.8% expected) and **under-selects D** (16.3% vs 26.5% expected). Accuracy collapses to **29.5%** when the correct answer is D, barely above the 25% random baseline.

Takeaways

- Language matters, but only for some models.** Gemini shows a robust English advantage of 2.4–10.0 pp; GPT-5.4-mini does not. Cross-lingual transfer is uneven across frontier models.
- MC format is not free.** Option-position bias can disadvantage some models even when their open-ended reasoning is fine.
- Validate before deploying.** Test candidate models on the **target language** and in the **intended evaluation format**. English-only, MC-only benchmarks can mislead.

References

Chen et al. (2024). Breaking language barriers in multilingual mathematical reasoning. *ACL*.
Cobbe et al. (2021). Training verifiers to solve math word problems. *arXiv:2110.14168*.
Hendrycks et al. (2021). Measuring mathematical problem solving with the MATH dataset. *NeurIPS*.
Myrzakhan et al. (2024). Open-LLM-Leaderboard: from MC to open-style questions. *arXiv:2406.07545*.
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